

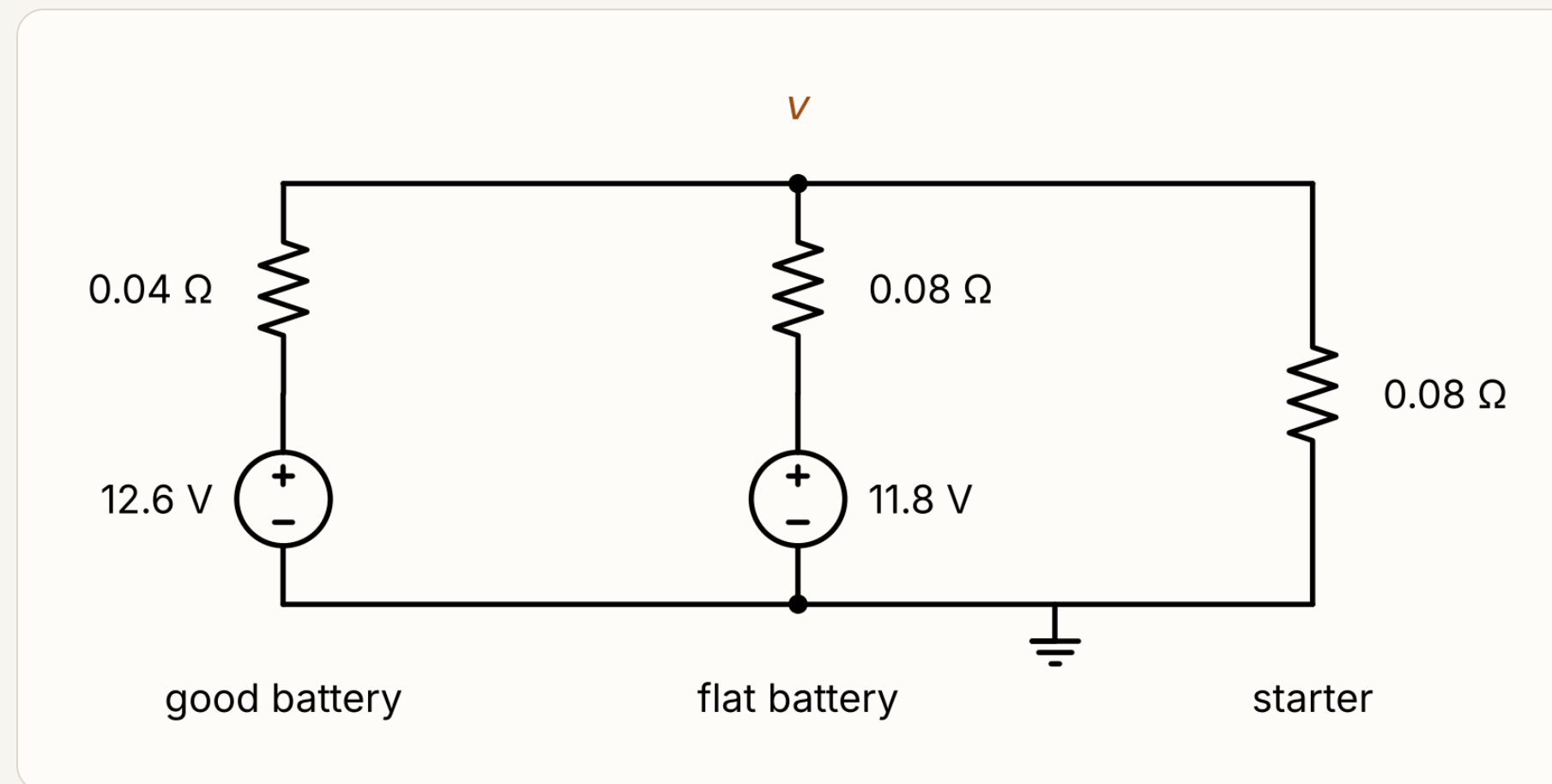
Node Voltages and Mesh Currents

Write exactly the equations a circuit needs: KCL at the essential nodes, or KVL around the meshes.

Prepared by Dr. Adel Daaboul

PREDICT FIRST

Jump-starting a car: while the starter cranks, does the flat battery help, or keep charging?



TODAY

By the end of Part 1 you can

1 • Count

Find the essential nodes and meshes, and say how many equations each method needs.

2 • Node voltages

Write KCL equations, with dependent sources, known node voltages and supernodes.

3 • Mesh currents

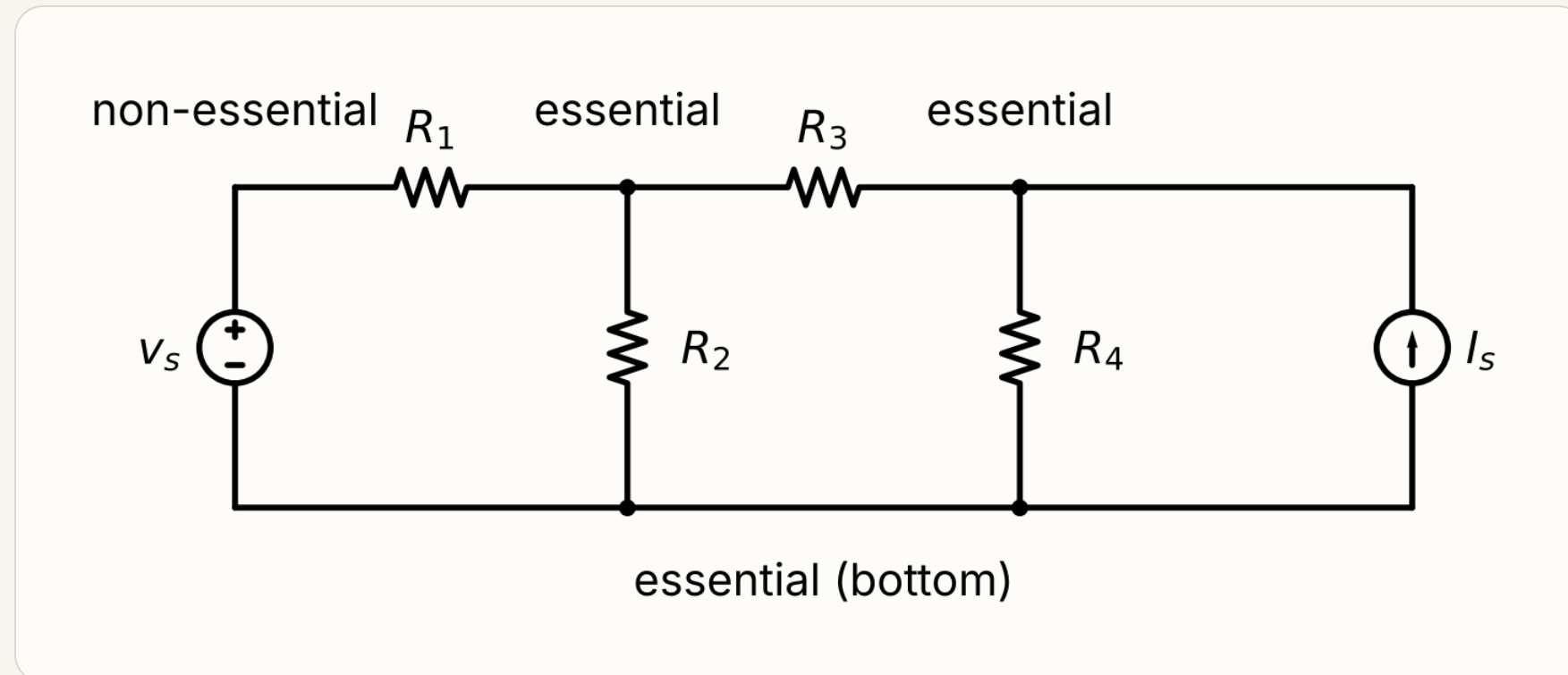
Write KVL equations, with dependent sources, known mesh currents and supermeshes.

4 • Choose and check

Pick the method with fewer equations, and check the answer without solving again.

COUNT BEFORE YOU WRITE

Essential nodes and branches set the count



Essential node: three or more elements meet.

Essential branch: a path between essential nodes that passes through no other essential node.

Mesh: a loop with no other loop inside it.

$$\text{node equations} = n_e - 1, \quad \text{mesh equations} = b_e - (n_e - 1)$$

Here $n_e = 3$ and $b_e = 5$: 2 node equations, or 3 mesh equations (I_s fixes one mesh current, so 2 to solve: slide 15).

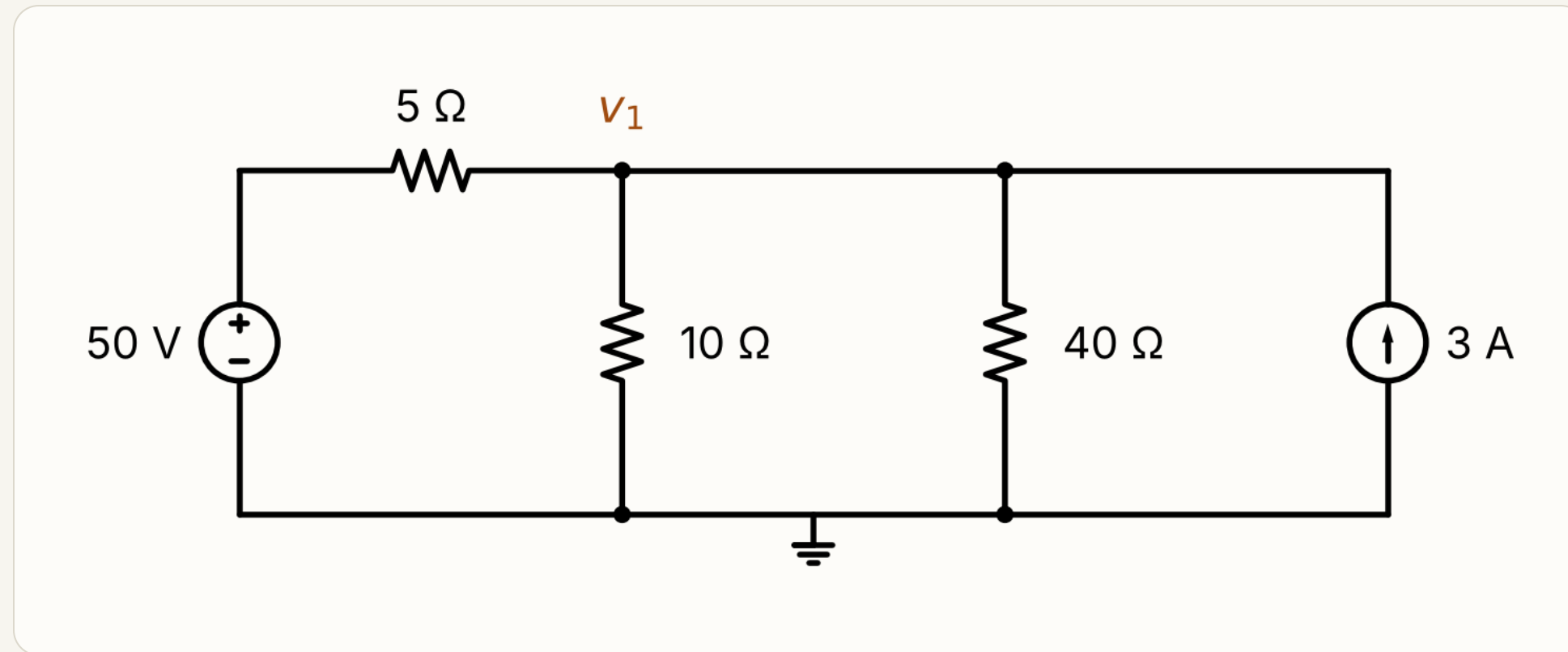
THE NODE-VOLTAGE METHOD

Five steps, always in the same order

- 1** Mark the essential nodes. Choose one as the reference, usually the one with the most branches.
- 2** Name the voltage of every other essential node, measured from the reference.
- 3** Write KCL at each node whose voltage is unknown (a supernode counts as one): currents **leaving** are positive.
- 4** Add one constraint for each dependent source, and for each supernode.
- 5** Solve, then substitute your numbers back into KCL at each node.

NILSSON EXAMPLE 4.3

One essential node besides the reference



KCL at node 1, currents leaving positive:

$$\frac{v_1 - 50}{5} + \frac{v_1}{10} + \frac{v_1}{40} - 3 = 0 \Rightarrow v_1 = 40 \text{ V}$$

NILSSON EXAMPLE 4.3

Branch currents and source powers from v_1

i_a flows left to right in the $5\ \Omega$; i_b and i_c flow down the $10\ \Omega$ and the $40\ \Omega$:

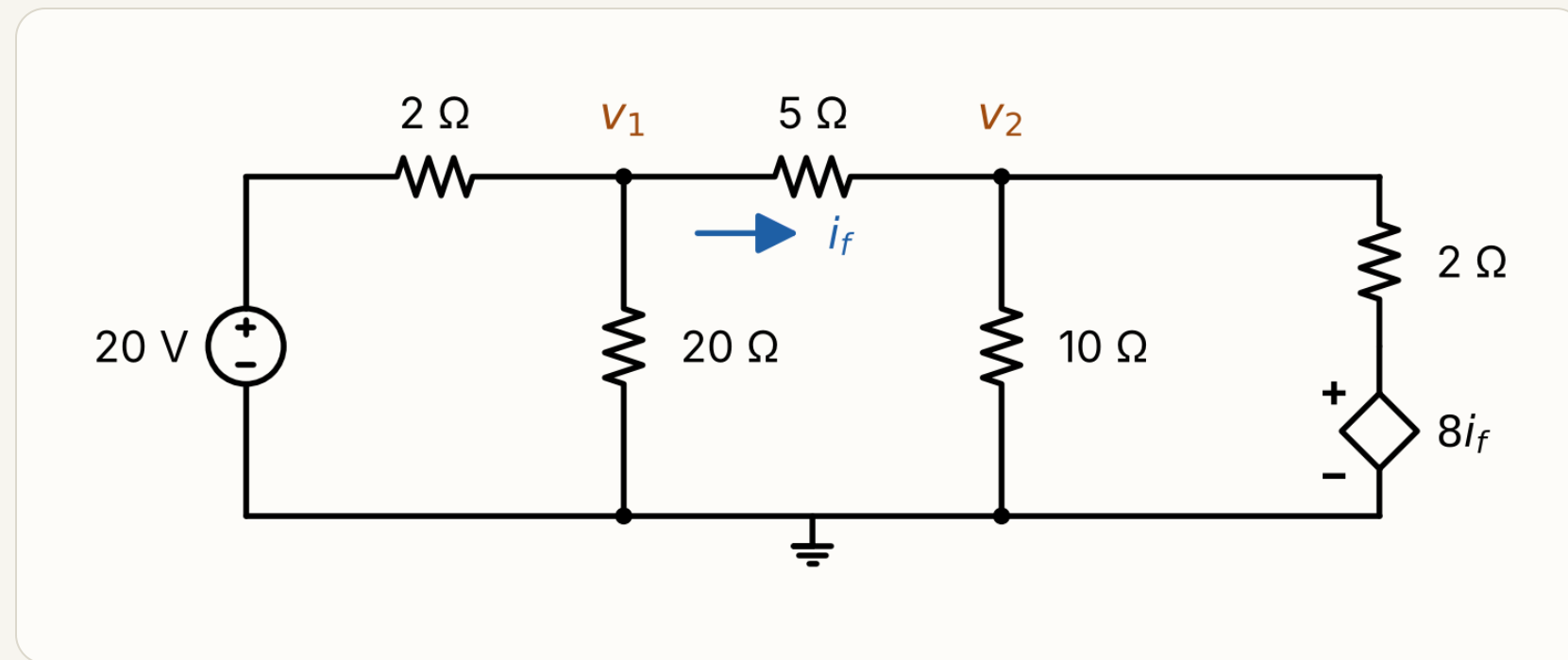
$$i_a = \frac{50 - 40}{5} = 2\ \text{A}, \quad i_b = \frac{40}{10} = 4\ \text{A}, \quad i_c = \frac{40}{40} = 1\ \text{A}$$

$$p_{50\ \text{V}} = -(50)(2) = -100\ \text{W}, \quad p_{3\ \text{A}} = -(40)(3) = -120\ \text{W}$$

Both sources deliver: 220 W in total. The resistors absorb $20 + 160 + 40 = 220\ \text{W}$.

NILSSON EXAMPLE 4.4

A dependent source adds one constraint equation



KCL at nodes 1 and 2:

$$\frac{v_1 - 20}{2} + \frac{v_1}{20} + \frac{v_1 - v_2}{5} = 0$$

$$\frac{v_2 - v_1}{5} + \frac{v_2}{10} + \frac{v_2 - 8i_f}{2} = 0$$

Constraint, from the figure:

$$i_f = \frac{v_1 - v_2}{5}$$

Write the controlling current i_f from the node voltages, then substitute.

NILSSON EXAMPLE 4.4

Solve, find the power, then check with your numbers

$$v_1 = 16 \text{ V}, \quad v_2 = 10 \text{ V}, \quad i_f = 1.2 \text{ A}$$

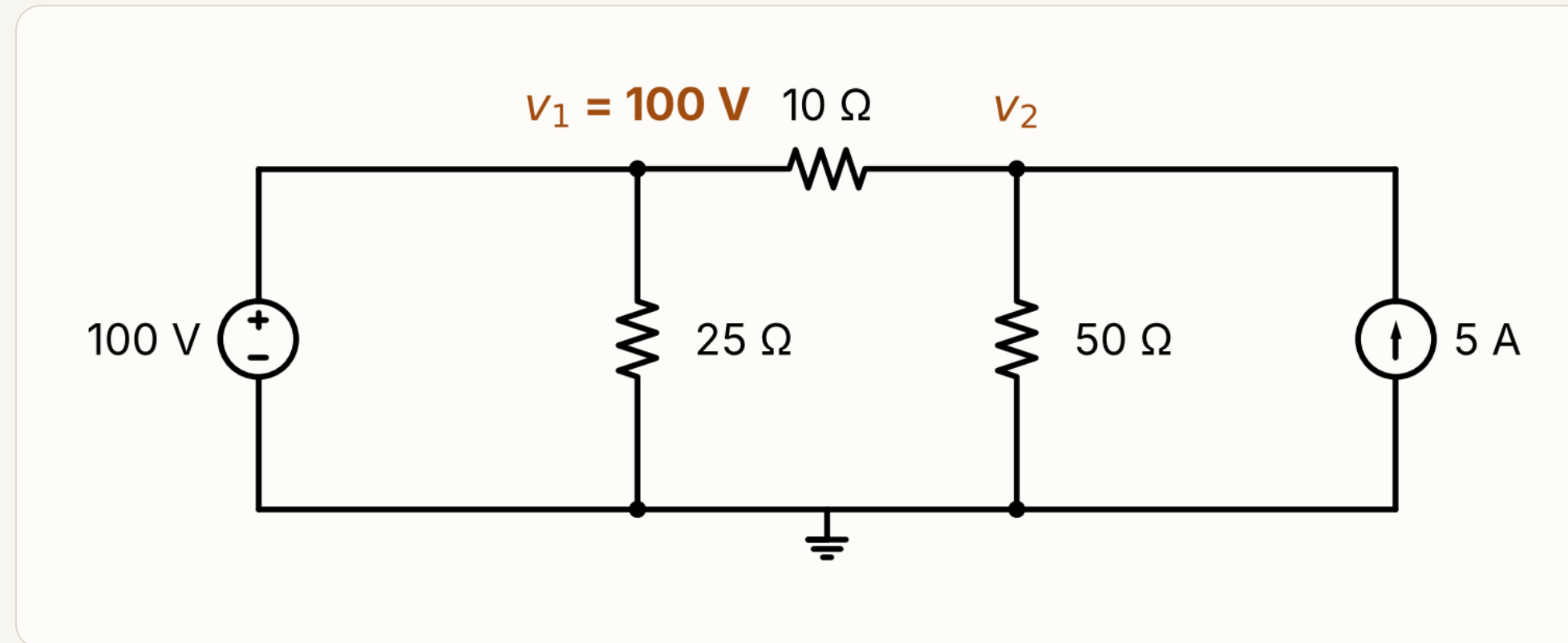
$$p_{5\Omega} = (5)(1.2)^2 = 7.2 \text{ W}$$

Check: substitute into KCL at node 1. The three currents must add to zero.

$$\frac{16 - 20}{2} + \frac{16}{20} + \frac{16 - 10}{5} = -2 + 0.8 + 1.2 = 0$$

A VOLTAGE SOURCE TO THE REFERENCE

It fixes a node voltage, so that node needs no equation

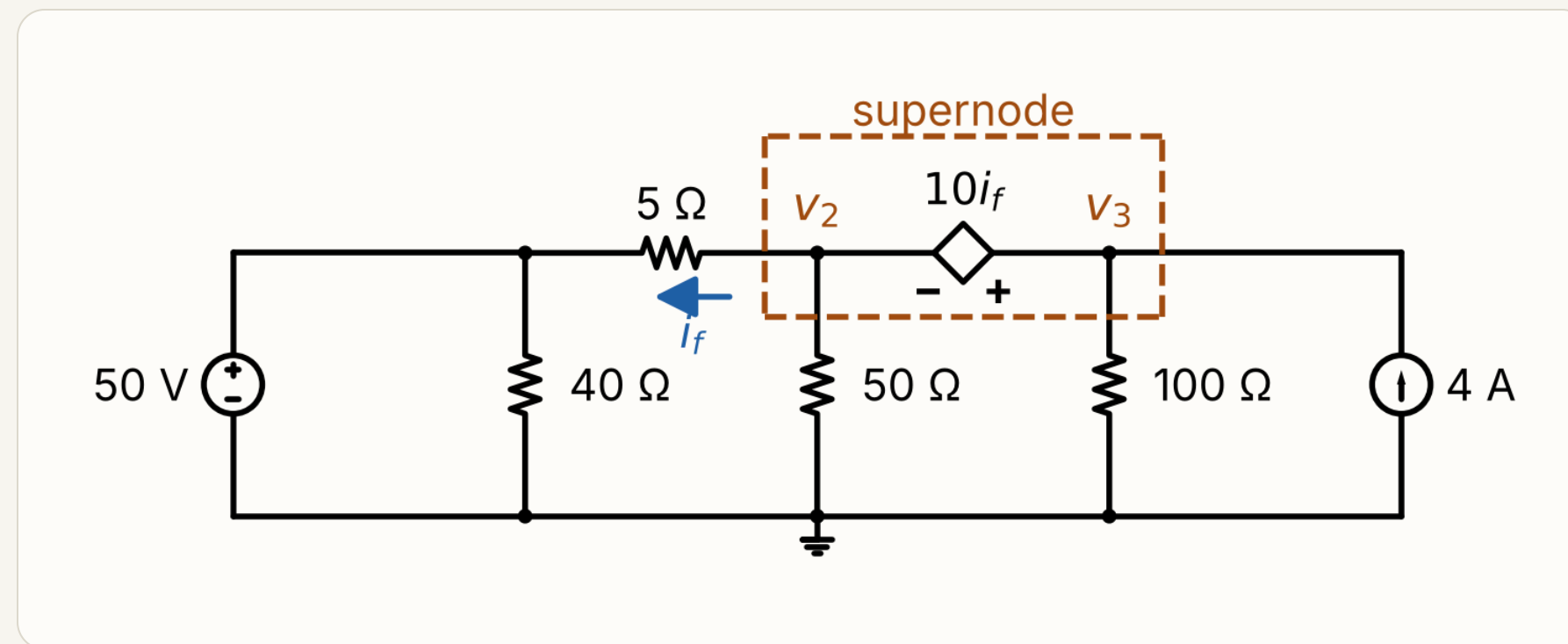


$$\frac{v_2 - 100}{10} + \frac{v_2}{50} - 5 = 0 \Rightarrow v_2 = 125 \text{ V}$$

The 25Ω resistor never appears: $v_1 = 100 \text{ V}$ is already known.

A SUPERNODE

A voltage source between two nodes: one KCL equation for both



KCL for the supernode:

$$\frac{v_2 - 50}{5} + \frac{v_2}{50} + \frac{v_3}{100} - 4 = 0$$

Constraint:

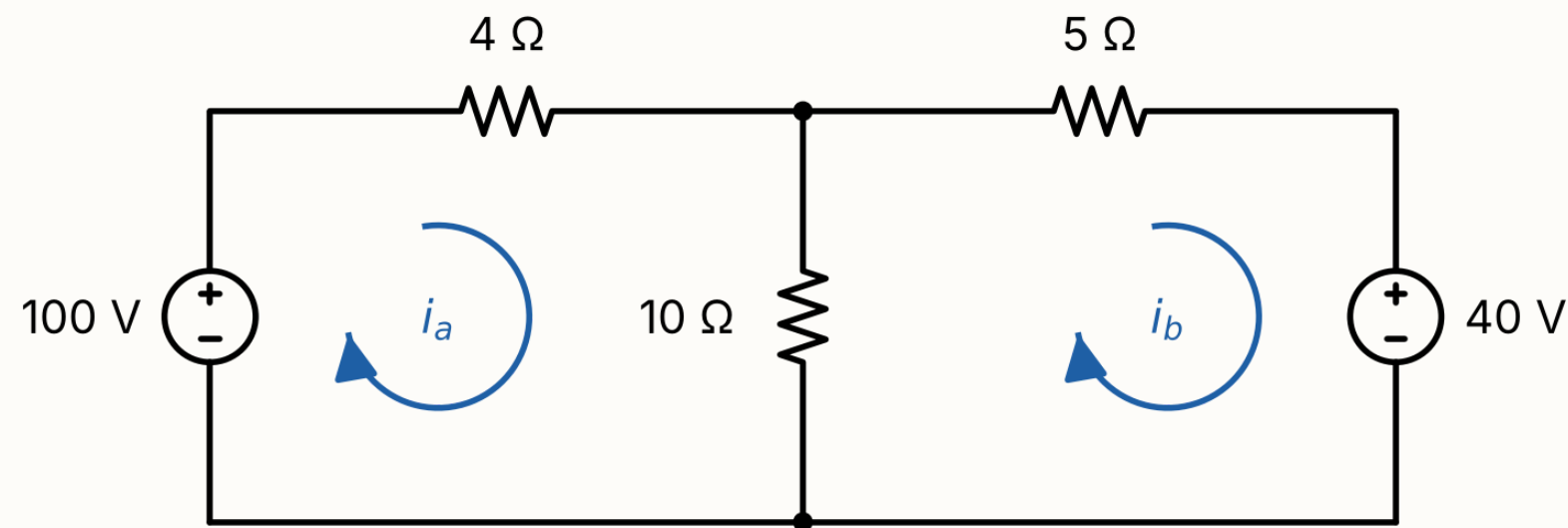
$$v_3 - v_2 = 10i_f, \quad i_f = \frac{v_2 - 50}{5}$$

Solve:

$$v_2 = 60 \text{ V}, \quad v_3 = 80 \text{ V}, \quad i_f = 2 \text{ A}$$

THE MESH-CURRENT METHOD

A clockwise current in each mesh; KVL around it, drops positive



Mesh a and mesh b:

$$-100 + 4i_a + 10(i_a - i_b) = 0$$

$$5i_b + 40 + 10(i_b - i_a) = 0$$

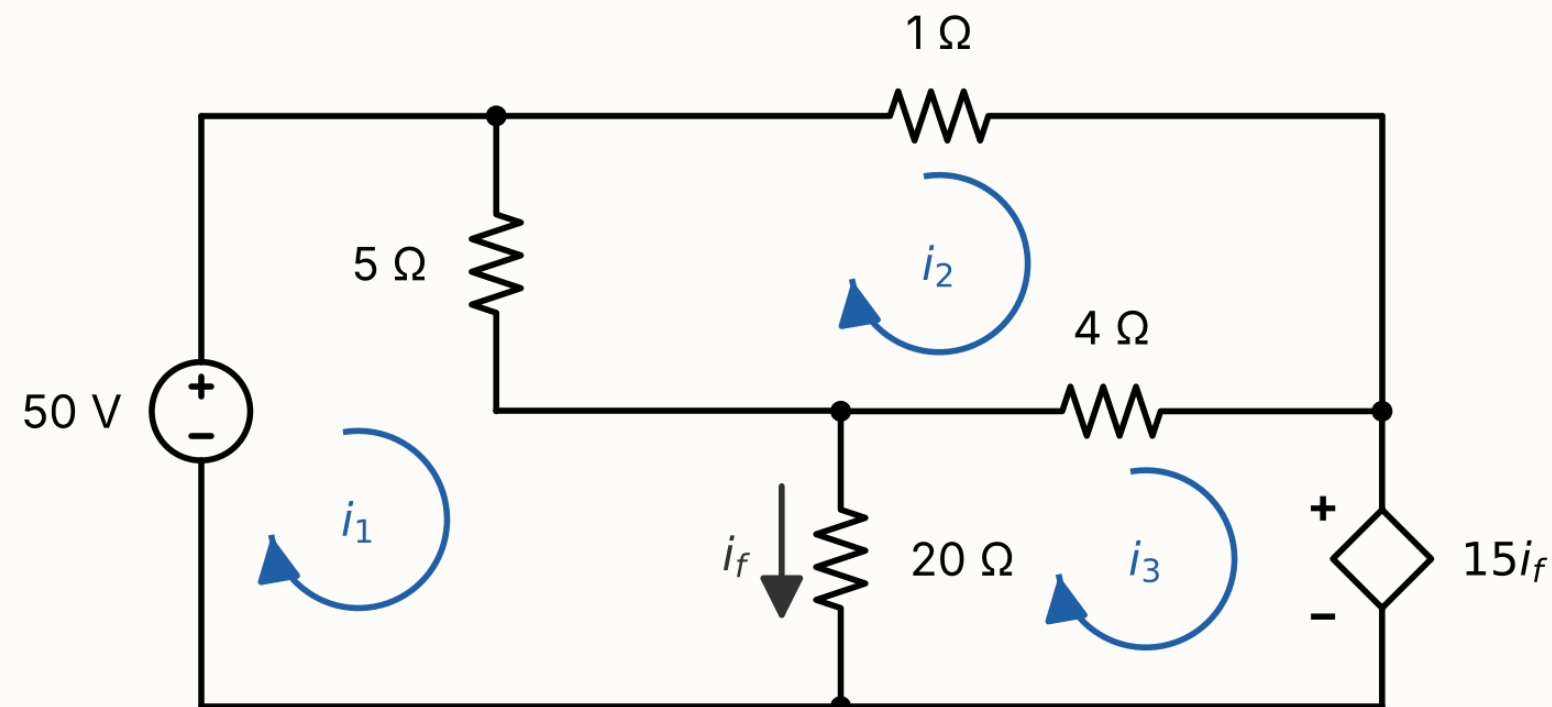
Solve:

$$i_a = 10 \text{ A}, \quad i_b = 4 \text{ A}, \quad i_{10\Omega} = i_a - i_b = 6 \text{ A}$$

A shared resistor carries the difference: in mesh b, write $10(i_b - i_a)$, your own mesh current first.

NILSSON EXAMPLE 4.7

Three meshes and a current-controlled voltage source



Meshes 1 and 2:

$$5(i_1 - i_2) + 20(i_1 - i_3) - 50 = 0$$

$$5(i_2 - i_1) + 1 i_2 + 4(i_2 - i_3) = 0$$

Mesh 3, and the constraint for the dependent source:

$$20(i_3 - i_1) + 4(i_3 - i_2) + 15 i_f = 0, \quad i_f = i_1 - i_3$$

NILSSON EXAMPLE 4.7

Solve, then find the power in the 4 Ω resistor

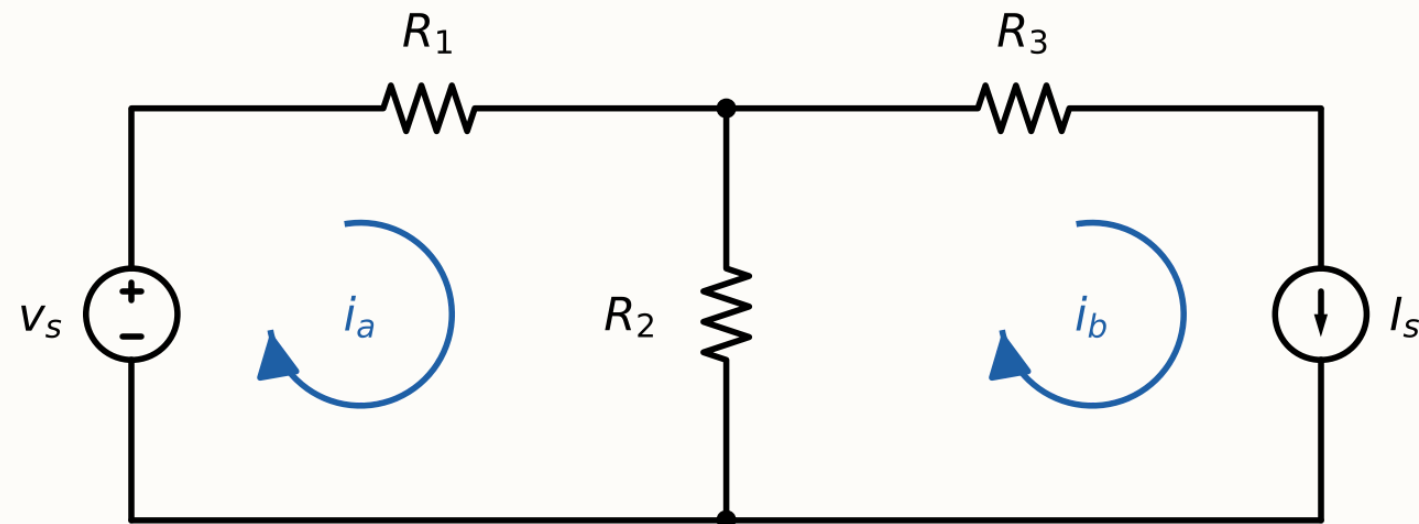
$$i_1 = 29.6 \text{ A}, \quad i_2 = 26 \text{ A}, \quad i_3 = 28 \text{ A}, \quad i_f = 1.6 \text{ A}$$

$$p_{4\Omega} = (4)(i_3 - i_2)^2 = (4)(2)^2 = 16 \text{ W}$$

Check mesh 2 with your numbers: $5(26 - 29.6) + 26 + 4(26 - 28) = -18 + 26 - 8 = 0$.

A CURRENT SOURCE IN ONE MESH

It fixes that mesh current: one equation fewer



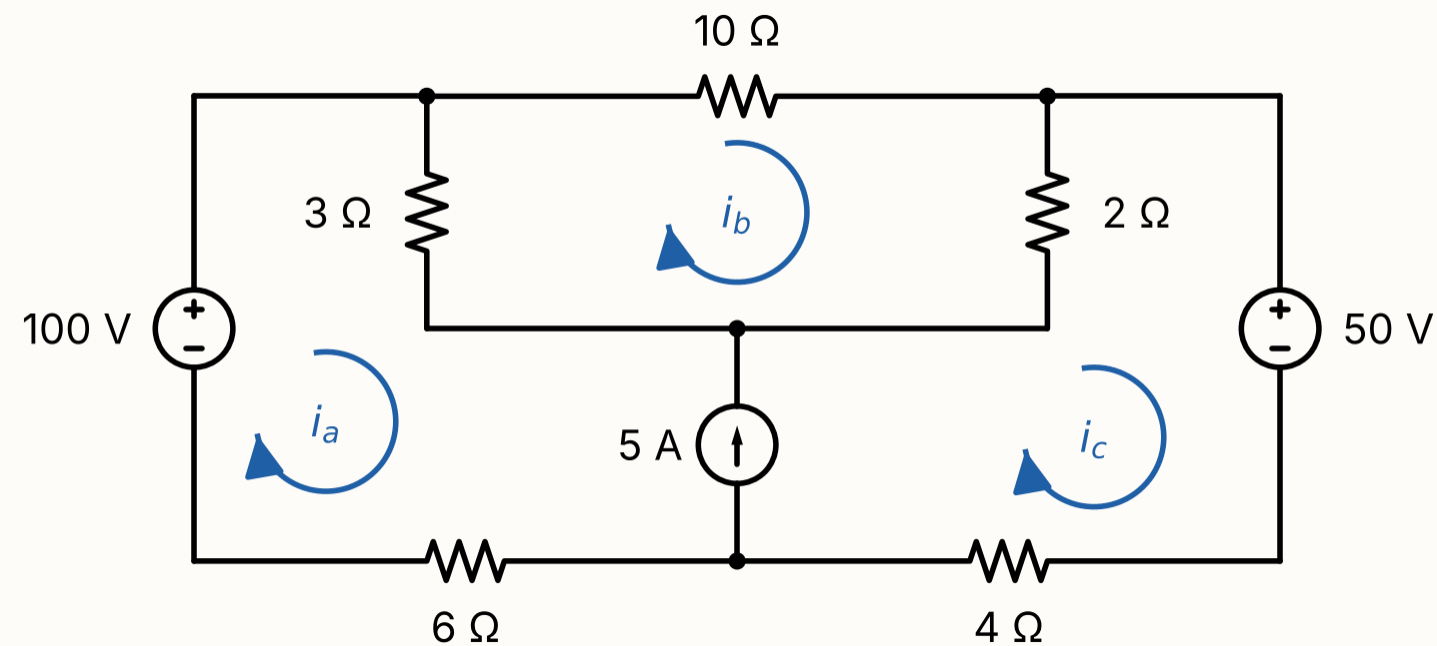
$$i_b = I_s \quad (\text{no KVL equation for mesh } b)$$

Write KVL only for mesh a.

Read the direction: here I_s points down the right side, the same way as the clockwise i_b . Pointing up, it would give $i_b = -I_s$.

A SUPERMESH

A shared current source: walk around both meshes



Supermesh: meshes a and c together, skipping the 5 A branch:

$$-100 + 3(i_a - i_b) + 2(i_c - i_b) + 50 + 4i_c + 6i_a = 0$$

Mesh b, and the constraint from the source:

$$10i_b + 2(i_b - i_c) + 3(i_b - i_a) = 0, \quad i_c - i_a = 5$$

Solve:

$$i_a = 1.75 \text{ A}, \quad i_b = 1.25 \text{ A}, \quad i_c = 6.75 \text{ A}$$

TWO METHODS, ONE PATTERN

Swap nodes for meshes, KCL for KVL, voltage sources for current sources

	Node voltages	Mesh currents
Unknowns	Node voltages	Mesh currents
Law	KCL, currents leaving positive	KVL, drops positive
Number of equations	$n_e - 1$	$b_e - (n_e - 1)$
Simplifying source	Voltage source: known node or supernode	Current source: known mesh or supermesh
Dependent source	One constraint equation	One constraint equation
Check	KCL at each node	KVL around an unused loop

CHOOSING A METHOD

Count the equations each method needs, then pick the smaller

Circuit	Node voltages	Mesh currents	Choose
Example 4.3 (slide 6)	1	2 (three meshes, one fixed by the 3 A)	Node
Fig. 4.19 (slide 12)	1	2	Node
Fig. 4.27 (slide 16)	3	2 (supermesh; the 5 A gives $i_c - i_a = 5$)	Mesh
Example 4.10 (book)	4 (a supernode)	6 (five meshes and a constraint)	Node

Fig. 4.19 with one node equation, v measured at the top of the $10\ \Omega$:

$$\frac{v - 100}{4} + \frac{v}{10} + \frac{v - 40}{5} = 0 \Rightarrow v = 60\text{ V}$$

Then $i_a = (100 - 60)/4 = 10\text{ A}$ and $i_b = (60 - 40)/5 = 4\text{ A}$, as on slide 12.

CHECKING AN AI ANSWER

An AI wrote mesh b of Fig. 4.19 like this. What is wrong?

$$5i_b + 40 + 10(i_a - i_b) = 0$$

Correct:

$$5i_b + 40 + 10(i_b - i_a) = 0$$

In mesh b the shared $10\ \Omega$ carries $i_b - i_a$. The slip gives $i_a = -30\text{ A}$ and $i_b = -52\text{ A}$; KVL around the outer loop then fails: $4(-30) + 5(-52) = -380$, but it must equal $100 - 40 = 60$.

In a 2025 study of Gemini 2.5 Pro on undergraduate circuit problems, misread source polarities caused about a third of its wrong answers, and misread current directions nearly as many.

CHECKING WITHOUT SOLVING AGAIN

Check with numbers, reading directions from the figure

Node answers

KCL at each node (Example 4.4, node 1).

$$\frac{16 - 20}{2} + \frac{16}{20} + \frac{16 - 10}{5} = -2 + 0.8 + 1.2 = 0$$

Mesh answers

KVL around a loop you did not write: the outer loop of Fig. 4.19.

$$-100 + 4(10) + 5(4) + 40 = 0$$

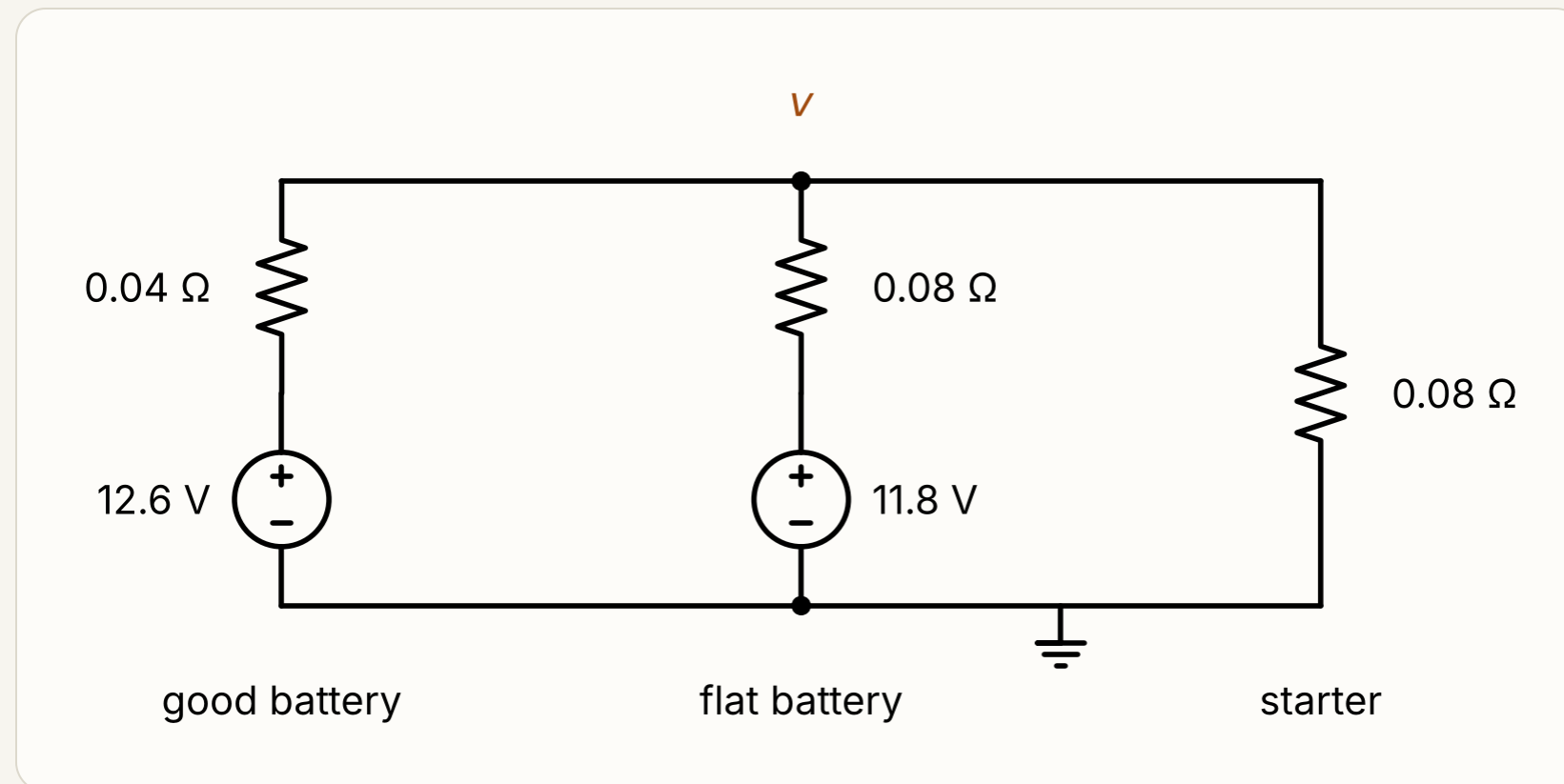
Either

A power balance over every element.

$$\sum p = 0$$

BACK TO THE JUMP START

One node equation answers the question



While cranking, KCL at the top node:

$$\frac{v - 12.6}{0.04} + \frac{v - 11.8}{0.08} + \frac{v}{0.08} = 0 \Rightarrow v = 9.25\text{ V}$$

Current out of the flat battery's + terminal:

$$i_{\text{flat}} = \frac{11.8 - 9.25}{0.08} = 31.9\text{ A}$$

Before cranking, without the starter, $v = 12.33\text{ V}$ and 6.67 A flows into the flat battery: it charges. While cranking, v falls below 11.8 V , so the flat battery helps with 31.9 A ; the good battery gives 83.8 A and the starter draws 115.6 A .

YOUR SELF-CHECK

Three habits for every node or mesh problem

Count first

Essential nodes and meshes;
spot supernodes, supermeshes
and known values.

Signs by rule

Currents leaving positive;
drops positive; your own mesh
current first.

Check with numbers

KCL at each node, KVL around
an unused loop, or a power
balance.

WHAT COMES NEXT

Practise openly, then show it on your own

In class

Booklet Problems 4.1 to 4.12.

At home

The 16-item practice sheet. AI tools allowed; not marked.

Week 7

15-minute quiz, no devices.
Every question is a twin of a practice-sheet question.

Part 2 (weeks 7–9): source transformations, superposition, Thévenin and Norton equivalents, maximum power transfer. Every one of them uses node or mesh equations.

CHECKLIST

You can do these without help

- Count the essential nodes and meshes, and say how many equations each method needs
- Write node-voltage equations, including a dependent source, a known node voltage and a supernode
- Write mesh-current equations, including a dependent source, a known mesh current and a supermesh
- Choose the method with fewer equations, and say why
- Find the power of every source and say whether it absorbs or delivers
- Check a solution with KCL at each node, KVL around an unused loop, or a power balance